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Single-Chip Radar Sensor



The reference design offers high RF performance with a compact design, supporting 4x4 MIMO, object detection over 200 metres, and high-speed data streaming, enabling faster development for safety and driving assistance applications.

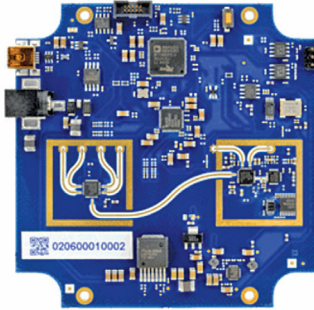
The Pluto radar sensor reference design from NXP Semiconductors features a 28nm single-chip radar SoC with four transmitters and four receivers, providing high RF performance and flexibility for corner and front radar applications. It integrates multiple processor cores and hardware accelerators for enhanced radar processing, outputting FFT data, point clouds, or tracked objects. It includes a patch antenna and supports edge and hybrid architecture capabilities to accelerate product development. With a compact PCB size of 50 x 50mm and sensor dimensions of 60 x 60 x 35mm, it features a 4-transmitter and 4-receiver antenna setup, enabling 4 x 4 MIMO and object detection beyond 200 metres. Supporting raw data capture, it enhances radar processing and analysis. The ready-to-use design accelerates product development by allowing companies to reuse schematics and hardware, providing a strong foundation for radar software development and reducing time to market.

The reference design suits safety features like cruise control, emergency braking, blind spot detection, and spotting obstacles and hazards.

NXP Semiconductors

<https://www.electronicsforu.com/electronics-projects/radar-reference-design>

24GHz Radar Sensor



The reference design offers quick deployment with minimal setup, supporting rapid prototyping for R&D. It includes hardware, software, and a complete RF signal chain, enabling efficient radar experiments and data analysis.

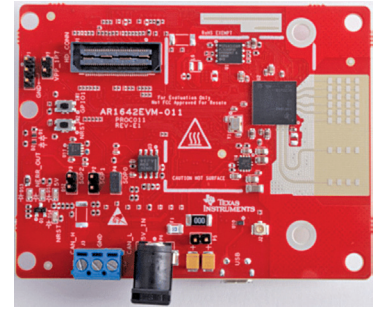
The Demorad platform is a 24GHz radar sensor reference design from Analog Devices that enables quick deployment with ready-to-use software examples and a simple setup process, allowing users to get the radar running in minutes. It supports rapid prototyping for R&D applications, helping users explore radar technology for detecting objects, tracking movement, and measuring angular position, velocity, and range. The platform includes hardware and software, streamlining the evaluation of 24GHz radar by reducing development time and technical complexity. It features a complete RF and analogue signal chain with DSP processing, providing an RF-to-bits solution for radar experiments with minimal setup time. Full software support includes a graphical user interface, DSP radar function libraries, and the ability to export raw data for post-processing using Matlab tools. Built on a single PCB with two transmitters and four receivers, the platform detects object range and velocity up to 200 metres, with a resolution of 75cm. Its field of view spans 120° in azimuth and 15° in elevation, with digital beamforming used to calculate angular information.

Applications include traffic monitoring, car parking, UAV/drone collision avoidance, altimeters, and security systems.

Analog Devices (ADI)

<https://www.electronicsforu.com/electronics-projects/24-ghz-radar-sensor-reference-design>

Short-Range Radar



The reference design offers upto 80 metres range, detecting 200 objects and tracking 24 at 90kph. It supports multi-mode radar, enhancing safety features like lane change assist and collision avoidance.

The TIDEP-0092 reference design from Texas Instruments is a short-range radar (SRR) solution built around the AWR1642 evaluation module (EVM), enabling estimating and tracking object positions and velocities within a range of up to 80 metres. Features such as self-parking, lane change assistance, adaptive cruise control, autonomous braking, and collision avoidance rely on radar sensors to detect and track nearby objects. The reference design can detect up to 200 objects within 80 metres and track 24 moving objects at speeds upto 90kph. Operating in multi-mode radar, it tracks objects at 80 metres while generating a detailed point cloud of objects within 20 metres. The design is a foundation for SRR automotive applications and can extend range beyond 80 meters with a higher-gain antenna. The AWR1642, a single-chip FMCW sensor in the 76 to 81GHz band, features four receivers, two transmitters, and a closed-loop phase-locked loop (PLL) for precise chirp synthesis. It supports non-MIMO and time-division multiplexed MIMO configurations, providing enhanced angle resolution with eight virtual receiver antennas.

The reference design is suitable for applications like lane change assist, autonomous parking, cross traffic alert, and blind spot detection, enhancing autonomous vehicle control by improving safety and quality of life.

Texas Instruments (TI)

<https://www.electronicsforu.com/electronics-projects/short-range-radar-reference-design>